

Garima Kumar

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EDUCATION

University of Illinois Chicago (UIC)

B.S in Data Science Chicago, IL

Dec 2025

GPA: 3.12/4.00

SKILLS

Languages: Python, C++, C, R, SAS, SQL, Julia | **Tools:** Pandas, NumPy, Scikit-learn, Coq, RefinedC, Github, Tableau, Power BI

Skills: Debugging and Testing, Descriptive and Inferential Statistics, Hypothesis Testing, ANOVA, AI, ML, Data Analysis, Data Mining, ETL, Regression, API, LLM

EXPERIENCE

Undergraduate Researcher | NSF-Funded Project on Formal Program Verification

Jan 2025 – Dec 2025

University of Illinois Chicago (UIC)

- Verified C programs using RefinedC and Coq, ensuring 100% memory and logic correctness
- Applied 15+ Coq tactics per proof to model control flow and memory usage

Research Aide- Technical | Internship

May 2025– Aug 2025

Argonne National Laboratory – Lemont, IL

LLM Slack Integration (Groq API)

- Engineered a Slack-integrated chatbot using Groq's LLaMA3-70B model, leveraging the Slack Bolt SDK and OpenAI-compatible APIs to automate research Q&A and streamline team communication workflows.
- Enabled intelligent, context-aware conversations by implementing threaded message handling and Slack's assistant.threads.setSuggestedPrompts, driving higher engagement and reduced response latency.

ORCID Publication Scraper

- Designed and deployed a robust ORCID publication scraper in Python, automating the extraction, parsing, and transformation of scientific metadata from JSON to clean CSV formats.
 - Accelerated large-scale data ingestion for author profiling across hundreds of ORCID IDs, boosting metadata accuracy and amplifying visibility of institutional research contributions.
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LEADERSHIP & MENTORSHIP

Teaching Assistant - CS 377: Ethics in Computer Science

Aug 2025 – Dec 2025

University of Illinois Chicago (UIC)

- Graded and evaluated 60+ upper-division student submissions weekly, applying standardized rubrics to ensure fairness and consistency in assessment.
- Delivered actionable written feedback that improved clarity and argument strength for 80%+ of students, reinforcing ethical reasoning skills in computing.

Engineering Success Program Mentor

Aug 2024 – Dec 2025

UIC College of Engineering

- Led and mentored **100+ first-year engineering students** in ENGR 100, supporting their academic transition.
 - Graded assignments with detailed feedback, improving student performance and understanding of course material.
 - Streamlined communication by managing the shared inbox and hosting weekly office hours, while enhancing engagement through Engineering Expo website updates.
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PROJECTS

Cell Type Classification (20k+ scRNA-seq samples)

Dec 2025

- Built ML pipeline with normalization and PCA (50 PCs, 90%+ variance) to classify 8 cell types from gene data
- Achieved >99% validation accuracy across Random Forest, Logistic Regression, and KNN using stratified, imbalance-aware training
- Improved minority-class F1 from ~0.90→0.96 via confusion-matrix-guided hyperparameter tuning

Project Bull vs. Bear

Apr 2025

- Modeled 100+ equities using moving averages and volatility indicators, revealing distinct cyclical vs. defensive sector behavior across market regimes
- Identified a +35% surge in trading volume during high-volatility phases, quantifying liquidity shifts during market stress
- Analyzed sector performance during the COVID-19 downturn, showing healthcare and consumer services recovered >20% faster than industrials and energy